

MICROBIOLOGY

CHAPTER 19

Microbial Taxonomy



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Reference:
Slides, Dr.Sereen
records,
Dr.Abdulkareem
lectures and the
book

Chapter 19: Microbial Taxonomy and the Evolution of Diversity

We're going to learn about: (●_●_●)

1. Defining taxonomy, natural classification & polyphasic taxonomy
2. Taxonomic ranks
3. Techniques of determining microbial taxonomy and phylogeny
4. Endosymbiosis Hypothesis\

🌸 Defining taxonomy, natural classification & polyphasic taxonomy

- Taxonomy: the science of biological classification (how to classify living things → in this particular case, bacteria)
 - It allows us to arrange a huge set of data and try to find relations or any logic to arrange them
 - Tax- means groups

منقدر ببساطة نحكي انو تخيل انك جاي على عالم لسا مو معروف فيه اي اشي عن البكتيريا وانت دورك انك تصنفهم ؛
التصنيف هاد يعني انك تحط كل مجموعة بنتشابه بشيء معين ضمن مسمى واحد وتشوف شو الصفة الي جمعت كل وحدة
فيهم مع الثانية وتسجل كل شي بحيث انو لما تلاقى اشي جديد تعرف مين هو او لوين تنسبه من المجموعات الي عندك

- taxonomy consists of three separate but interrelated parts:
 - **Classification:** arranging organisms into groups (taxa: groups| taxon: one group)
 - **Nomenclature:** the branch of taxonomy concerned with the assignment of names to taxonomic groups in agreement with published rules
 - **Identification:** the practical side of taxonomy—the process of determining if a particular isolate belongs to a recognized taxon and, if so, which one
 - ❖ When you want to classify bacteria, you always have to start with a pure culture (ch7-page 16)
- Natural Classification: One of the oldest classification systems that arranges organisms into groups whose members share many characteristics.
→ Carl von Linné, or Carolus Linnaeus as he often is called, developed the first natural classification in the middle of the 18th century
 - It was based on **anatomical** characteristics
 - Disadvantage: it does not necessarily provide information on evolutionary relatedness

- Polyphasic Taxonomy: it is used to determine the **genus** and **species** of a newly discovered prokaryote
→ As the term "polyphasic" suggests, there are many aspects to describe the microorganism. These include phenotypic, phylogenetic (i.e., the evolutionary history), and genotypic features.

Table 19.1 Components of Polyphasic Taxonomy			
Classification System	Basis of Classification	Traits	Microbial Example
Phenetic	Phenotype	Morphology Physiology	<i>Clostridium</i> spp. (anaerobic, endospore-forming, Gram-positive rod) compared to <i>Bacillus</i> spp. (aerobic, endospore-forming, Gram-positive rod)
Genotypic	Genes and genomes	DNA:DNA hybridization Average nucleotide identity	<i>E. coli</i> strain K12 compared to <i>E. coli</i> O157:H7 <i>E. coli</i> compare to <i>Salmonella enterica</i>
Phylogenetic	Evolution	rRNA genes Conserved protein-coding gene	Archaea compared to Bacteria

ال Phenetic classification يعتمد بشكل اساسي على ال phenotype او "الطراز الشكلي" (الي هو نتاج ال expression للجينات) وعشان نعملو منطع على ال morphology وال physiology.

ال Genotypic يعتمد على الجينات تبعت البكتيريا او بشكل عام الجينوم تبعها. نعمل DNA:DNA hybridization أو/و Average nucleotide identity

ال Phylogenetic اساسو ال evolution. يعتمد على ال rRNA وال Conserved protein-coding genes عشان يشوفوا اذا في علاقة تطورية بين نوعين من البكتيريا شوفوا الامثلة عليهم من جدول شو الانواع الي قارنوها ببعض بناء على كل نوع من ال classification وهلا رح نشرح كل وحدة بالتفصيل (—)

- Phenetic Classification:

- Groups organisms together based on mutual similarity of phenotypes
- This system succeeded in bringing order to biological diversity and clarified the function of morphological structures.

هاد ال classification بيّن كيف ممكن الشكل يرتبط بوظيفة معينة. كمثال، لاحظوا انو امتلاك البكتيريا flagella والقدرة على اداء حركة معينة مرتبطين ببعض ففقدوا يعرفوا انو ال flagella (morphological structure) بتسبب على الاقل نوع حركة واحد (وهاد هو ال function)

- Can reveal evolutionary relationships, but not dependent on phylogenetic analysis

ممكن يقدر يبين بعض العلاقات التطورية بس مو زي ال phylogenetic وما يعتمد عليه
→ - i.e., doesn't weigh characters
ما بقدر يحدد اي صفة من صفات البكتيريا مهمة اكثر من الثانية للتفريق بين ال species
← عشان هيك منستخدم ال genotypic classification

- Best systems compare as many attributes as possible

- Genotypic Classification

- It compares the genetic similarity between organisms
- - individual genes or whole genomes can be compared
يعني يا اما منقارن جينات ببعضها او الجينوم كامل (كل المادة الوراثية)
- 70% homology → belong to the same species
⇒ However, there is now a consensus building to replace this metric with a genomics-based assay that measures the average nucleotide identity between organisms (رح نشرحه)

- Phylogenetic Classification

→ Also called Phyletic classification systems

Phylogeny : evolutionary development of a species

- Usually based on direct comparison of genetic material and gene products
→ Woese and Fox proposed using small subunit (SSU) rRNA nucleotide sequences to assess evolutionary relatedness of organisms
 - ❖ The small subunit is highly conserved because it's not highly exposed to mutations

عشان يعرفوا شو علاقة البكتيريا ببعضها تطورياً، بقارنوا ال small subunit rRNA بينهم لانو مو معرّض للطفرات كثير بالتالي ما صار عليه تغييرات كثير عبر الزمن.. ف البكتيريا القريبة من بعضها رح يكون في تشابه بال SSU rRNA بينهم

🌸 Taxonomic ranks

• Taxonomic Ranks Provide an Organizational Framework

- Microbes are placed in hierarchical taxonomic levels with each level or rank sharing a common set of specific features

يتم وضع ال microbes بمستويات مرتبة بتسلسل هرمي معين ويكون غير متداخل وكل مستوى ما يكون بس عنده الصفات تبعون الترتيب الي فوقه، بس برضو يكون عندو كمان صفات بتحدده اكثر

- Highest rank is domain

- Bacteria and Archaea - microbes only
- Eukarya - microbes and macroorganisms

- We have with 5 kingdoms :

Monera, Animalia, Plantae, Fungi and Protista

(Monera: only prokaryotes; Protista , Fungi, Plant and Animal kingdoms are all eukaryotes)

→ Bacteria and Archaea belong to Monera kingdom

→ Within domain we have:

Phylum → class → order → family → genus → species epithet, some microbes have subspecies

زمان ما كان في اشي اسمو domain.. كان أعلى rank هو ال kingdom بس هال domain صار اعلى من ال kingdom وبحتوي على ال eukarya وال bacteria وال archaea لهيك ممكن تلاقوا بكتب انهم حاطين بس kingdoms او بس domains او التنين مع بعض.. حسب

الـphylum فيه مجموعة من الـclasses

والـclass فيه مجموعة من الـorders

والـorder فيه مجموعة من الـfamilies

والـfamily فيها مجموعة من الـgenera

والـgenus مجموعة من الـspecies واحيانا يكون في ضمن الـspecies الوحدة مجموعة من الـsubspecies

خلينا على سبيل المثال نتبع الـHierarchical Taxonomy للـShigella :-

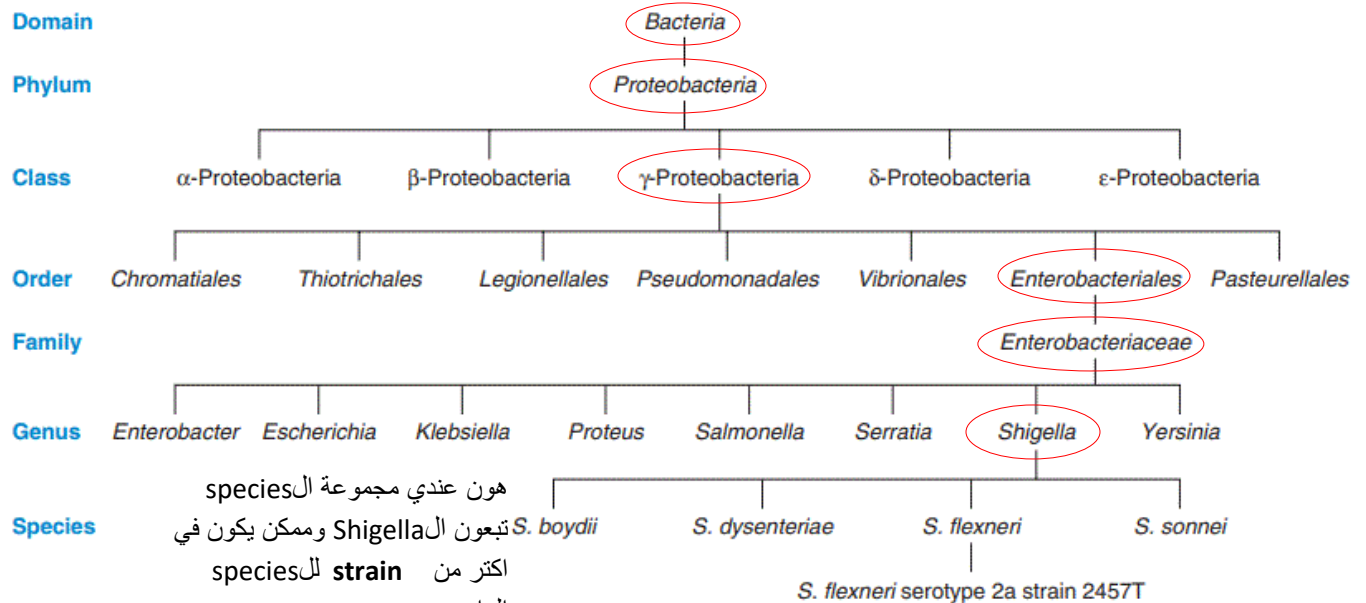


Figure 19.1 Hierarchical Arrangement in Taxonomy. In this example, members of the genus *Shigella* are placed within higher taxonomic ranks. Not all classification possibilities are given for each rank to simplify the diagram. Note that *-ales* denotes order and *-ceae* indicates family.

- Genus:
 - A genus is a well-defined group of one or more species that is clearly separate from other genera.
 - Often disagreement among taxonomists about the assignment of a specific species to a genus
- Species
 - Definition 1: collection of strains that share many stable properties and differ significantly from other groups of strains
 - Definition 2: collection of organisms that share the same sequences in their core housekeeping genes
- Strains
 - It consists of the descendants of a single, pure microbial culture
 - Vary from each other in many ways:
 - **Biovars** (Biological variation) - differ biochemically and physiologically (like the ability to do fermentation)
 - **Morphovars** (Morphological variation)- differ in morphology
 - **Serovars** (Serological variation) - differ in distinctive antigenic (immunologically reactive) properties

***Serology:** Studying kinds of antibodies secreted when the body is attacked by bacteria or viruses

جسمنا يحتوي على **antibodies** معينة وهي الـ **antibodies** كل واحد منها يستجيب لـ **antigen** معين جاي من برا الجسم.. الـ **antigens** يكونوا على سطح البكتيريا.. ليهيك لما نقارن كيف كل وحدة منهم كيف بتأثر على الجسم وكيف ممكن يستجيب الجسم لكل وحدة منهم بناء على اختلاف الـ **antigens** بينهم.. بقدر افرق بين الـ **strains** المختلفة.

مثال على اختلاف الـ **strains** انو ممكن شخصين يكون عندهم *Helicobacter pylori* بس مو الاتنين عندهم **ulcer** (قرحة بالمعدة).
بكون واحد فيهم عندو **strain** معينة بتسبب الـ **ulcer** والثاني الـ **strain** الي عندو ما بتسبب. قدرة الـ **strain** الاولى انها تسبب **ulcer** هو امتلاكها **virulence factors** (factors that allow the bacteria to cause diseases)

- **Type Strain:** one of the strains that is usually one of the first strains of species studied and often most fully characterized
 - Not necessarily most representative member of species
 - Only those strains very similar to the type strain or type species are included in a species
- **بالعادة لما نوصل لعند الـ strain بتكون البكتيريا تم تعريفها بشكل كامل من حيث الـ morphology والـ serology والـ biochemistry وكل اشئ وما ضل اشئ ما حددناه**

- **Binomial System of Nomenclature:**

- Devised by Carl von Linné (Carolus Linnaeus)
(نفسو تبع الـ natural classification)
- Each organism has two names:
 - Genus name : italicized (مائل) and capitalized (e.g., *Escherichia*)
 - Species epithet : italicized but not capitalized (e.g., *coli*)
- Can be abbreviated after first use (e.g., *E. coli*)

لما بدنا نكتب اسم اي كائن على الارض بدنا نكتب اسم الـ genus تبعو والـ species مثلا احنا اسمنا *Homo Sapiens*؛
homo هو الـ genus تبعنا و *sapiens* هو الـ species .

ودايما لما نكتب الاسم العلمي لكائن لازم يكون مائل (الـ genus والـ species) والـ Genus فقط لازم يكون اول حرف الو

كبير ← *Escherichia coli* وبعد اول استخدام مختصر الـ genus ف بتصير *E.coli*

❖ A new species cannot be recognized until it has been published in the *International Journal of Systematic and Evolutionary Microbiology*

- The species name is stable; the oldest epithet for a particular organism takes precedence and must be used. In contrast, a generic name can change if the organism is assigned to another genus. For example, some members of the genus *Streptococcus* were placed into two new genera, *Enterococcus* and *Lactococcus*, based on rRNA analysis and other characteristics. Thus *Streptococcus faecalis* is now *Enterococcus faecalis*.

🌸 Techniques for determining microbial taxonomy and phylogeny

Many different approaches are used in classifying and identifying microorganisms that have been isolated and grown in pure culture. These approaches are divided into two groups: Classical & Molecular approaches

A. Classical approaches

morphological - physiological -
biochemical - ecological and genetic

1. Morphological Characteristics

- Morphological features are important in microbial taxonomy for many reasons:
 - Morphology is easy to study and analyze (see table 19.2)
 - morphological comparisons are valuable because structural features depend on the expression of many genes and are usually genetically stable. Thus morphological similarity often is a good indication of phylogenetic relatedness.

2. Physiological and Metabolic Characteristics

- Physiological and metabolic characteristics are very useful because they are directly related to the nature and activity of microbial proteins

يعني مثلا اذا منتبج النواتج النهائية لعملية ال fermentation بنوع جديد
مكتشف.. ممكن هدول النواتج يكشفوا عن وجود metabolic enzymes
معينة والجينات الي انتجت هدول الانزيمات ..لهيك تحليل مثل هاي الصفات
الي بتشبه ال fermentation (مثل ال energy metabolism and
microbial nutrient transport) بتعطيني تفريق مباشر بين ال
see table 19.3. genomes

Table 19.2	Some Morphological Features Used in Classification and Identification
Feature	Microbial Groups
Cell shape	All major groups ¹
Cell size	All major groups
Colonial morphology	All major groups
Ultrastructural characteristics	All major groups
Staining behavior	Bacteria, some fungi
Cilia and flagella	All major groups
Mechanism of motility	Gliding bacteria, spirochetes, protists
Endospore shape and location	Some Gram-positive bacteria
Spore morphology and location	Bacteria, protists, fungi
Cellular inclusions	All major groups
Colony color	All major groups

Table 19.3	Some Physiological and Metabolic Characteristics Used in Classification and Identification
Carbon and nitrogen sources	
Cell wall constituents	
Energy sources	
Fermentation products	
General nutritional type	
Growth temperature optimum and range	
Motility	
Osmotic tolerance	
Oxygen relationships	
pH optimum and growth range	
Photosynthetic pigments	
Salt requirements and tolerance	
Secondary metabolites formed	
Storage inclusions	

3. Biochemical Characteristics

2 important techniques: (الدكتور والدكتورة ما ذكروهم ولكن موجودين بالكتاب)

- fatty acid methyl ester (FAME) analysis: reveals specific differences in chain length, degree of saturation, branched chains, and hydroxyl groups
- mass spectrometry (MS): resulted in the fast and accurate identification of bacteria based on the presence of specific, highly abundant proteins.
→ the specific type is called matrix-assisted laser desorption/ionization-time of flight (MALDI-ToF) and it enables the analysis of complex biomolecules that could not previously be studied by MS.

4. Ecological Characteristics

الخصائص البيئية كثير مهمة لدراسة اي عينة من البكتيريا ومعرفة المصدر الي جاي منو البكتيريا (بيئتها) بدلني على اشيء كثير.. مثلا خلتنا ناخذ البكتيريا الي يتسبب التهاب الحلق.. معرفة بيئتها او هي وين بتعيش خلتنا نعرف انو درجة الحرارة المثالية لنموها هي 37° (درجة حرارة اجسامنا)
كمان مثال لما ندرس البكتيريا الي جاي من البحر الميت.. اول ما عرفنا انها جاي من هناك عرفنا انو اكيد لازم ننميتها على agar عندو تركيز مشبع من الاملاح
معرفة البيئة تبعت البكتيريا مو بس بتخلينا نعرف كيف ننميتها بالمختبر بس برضو بتعطينا فكرة عن كيف ممكن ننقلها من بيئتها الاصلية للمختبر تبعا

- These characteristics are :
 - Life-cycle patterns
 - Symbiotic relationships : like those found between some kinds of bacteria and plants where bacteria take up nitrogen from the environment and give it to the plant
 - Ability to cause disease
 - Habitat preferences such as requirements for temperature, pH, oxygen, and osmotic concentration
 - Growth requirements

B. Molecular Approaches

- They are extremely important because almost no fossil record was left by microbes
- Allows for the collection of a large and accurate data set from many organisms
- Phylogenetic inferences based on these provide the best analysis of microbial evolution currently available

المolecular approaches كثير مهمين لأنو على عكس الكائنات الثانية الي الهم احافير ممكن تساعد في معرفة العلاقات التطورية بينهم.. الmicrobes ما عندهم ف. بلجنوا لهدول الapproches عشان يقدرنا يعرفوهم لأنهم بوفروا كمية كبيرة ودقيقة من الdata.

- Molecular characteristics for identification and classification:

1. nucleic acid base composition
2. nucleic acid hybridization
3. nucleic acid sequencing
4. genomic fingerprinting

1. Nucleic acid base composition

Nucleic bases are: Thymine (T), Cytosine (C), Guanine (G), Adenine (A)

Pyrimidines :

Each
composed of
One ring

Purines :

Each
compose
d of Two
rings

→ A binds with T by Two hydrogen bonds

→ C binds with G by Three hydrogen bonds

- We need heat to break nucleic acid bonds and convert the Double stranded DNA to a Single stranded DNA

→ The amount of heat is determined by the G-C component of nucleic acids because we need higher heat to break the three hydrogen bonds of G & C than the two bonds between T & A

لأن الـ G والـ C في بينهم 3 روابط هيدروجينية والـ A والـ T في بينهم رابطتين .. ف احنا منعتمد على كمية الـ G والـ C الموجودة بالـ DNA لنحدد الحرارة اللازمة لفصل سلاسل الـ DNA عن بعض لأنهم يحتاجوا حرارة اعلى
كيفية حساب الـ G+C Content:

$$\text{Mole\% } G+C = \frac{G+C}{G+C+A+T} \times 100$$

- G + C content can be determined from the melting temperature (T_m) of DNA
→ strands of DNA with a greater G + C content separate at higher temperatures; that is, the DNA has a higher melting temperature.

temperatures; that is, the DNA has a higher melting temperature.

For example, compare *Mycoplasma hominis*, with a mol% G + C of about 29 and a T_m of 65°C, with *Micrococcus luteus*, which is almost 79% G + C with a T_m of 85°C.

لاحظوا كيف لما ارتفعت الـ mol% G+C ارتفعت معها الـ T_m

- When a DNA sample is slowly heated, the absorbance increases as hydrogen bonds are broken and reaches a plateau when all the DNA has become single-stranded (figure 19.2). The midpoint of the rising curve gives the melting temperature
- If two organisms differ in their G + C content by more than about 10%, their genomes have quite different base sequences, indicating that they are not closely related and does not belong to the same genus

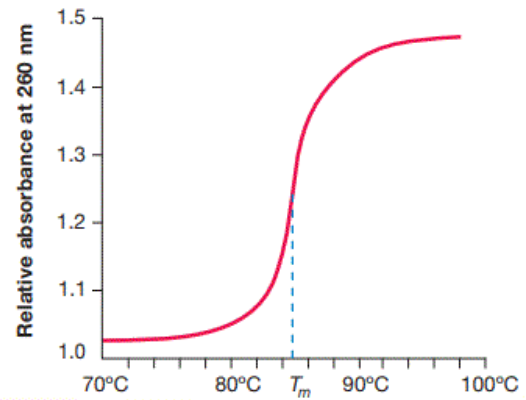


Figure 19.2 A DNA Melting Curve. The T_m is indicated.

MICRO INQUIRY Would this curve be shifted to the left or the right for a microbe with an exceptionally low G + C composition? Explain your answer.

عشان يكونوا نوعين من البكتيريا بنتموا لنفس ال genus.. لازم الاختلاف بال G+C content ما يزيد عن 10%

2. Nucleic Acid Hybridization

- DNA-DNA hybridization (DDH): measure of sequence homology
→ Hybrid: Two sources of One thing

Two bacterial cells :
Control & test

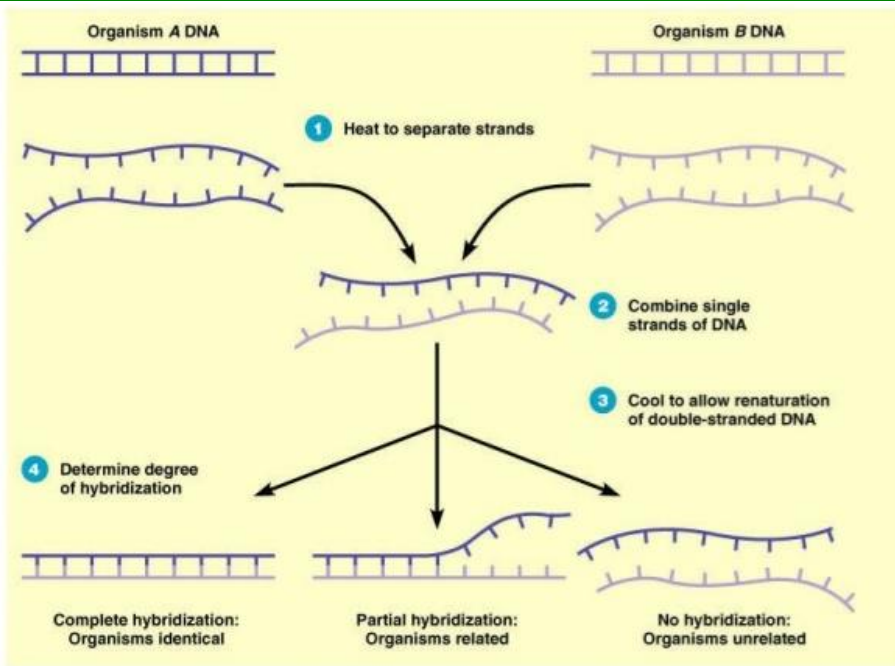
DNA

ال control هي الي بدي اقارن معها
وبتكون معروفة

ال test هي الي بدي اعملها
identification ولسا ما منعرفها
واللي منكون عارفين انها قريبة من
ال control ليهيك منقارنهم ببعض

- ✦ When a double stranded DNA(dsDNA) is heated, it will disassociate into a single stranded DNA(ssDNA) and when its cooled down it will reassociate

- Common procedure
 - bind nonradioactive DNA to nitrocellulose filter
 - incubate filter with radioactive single-stranded DNA
 - measure amount of radioactive DNA attached to filter



كيفية عمل DNA-DNA hybridization :

1. منجيب two organisms : test وال control بكتيريا
2. منفصل ال dsDNA تبع ال control برفع درجة الحرارة فيتحول ل 2ssDNA ومنحطهم على ال nitrocellulose filter ومنعملهم labeling باستخدام radioactive isotops
3. منفصل ال dsDNA تبع ال test ومنحط ال 2ssDNA molecules على ال nitrocellulose filter كمان
4. منحطهم على درجة حرارة 25°C عشان اذا في تسلسلات متشابهة بيناتهم يرجعوا

يرتبطوا مع بعض (reassociate) (ال noncomplementary strands رح يضلوا مو مرتبطين)

5. منعمل washing عشان نتخلص من اي ssDNA ما ارتبط بواحد ثاني وبرضو عشان نتخلص من ال excess isotopes

6. باستخدام جهاز اسمو **Beta Scintillation counter** منقيس اديه ضل isotopes (بعطيني رقم معين)

7. بعمل نفس الخطوات ولكن لل dsDNA تبع ال control مع نفسو عشان اقدر استخدمو كمرجع

8. بقرن الرقمين الي طلوعوا معي ب 6 و 7 وعلى اساسهم بقدر اعرف اذا ال test الها علاقة بال control وكيف ممكن اصنفها

- Two **strains** whose DNAs show at least 70% relatedness under optimal experimental hybridization conditions and less than 5% difference in T_m , often, but not always, are considered members of the same **species**
- an alternate approach has been sought called Average nucleotide identity (ANI)

3. Nucleic Acid Sequencing

→ The rRNAs from small ribosomal subunits (16S from bacterial and archaeal cells and 18S from eukaryotes) have become the molecules of choice for inferring microbial phylogenies and making taxonomic assignments at the genus level.

- The small subunit rRNAs (SSU rRNAs) are almost ideal for studies of:
 - 1- microbial evolution
 - 2- Relatedness
 - 3- genus identification for several important reasons:
 - they play the same role in all microorganisms
 - because ribosomes are absolutely necessary for survival and ribosomes cannot function without SSU rRNAs, the genes encoding these rRNAs cannot tolerate large mutations. Thus these genes change very slowly with time

- rRNA genes do not appear to be subject to horizontal gene transfer, an important factor in comparing sequences for phylogenetic purposes
- Comparative analysis of SSU rRNA sequences from thousands of organisms has demonstrated:
 - oligonucleotide signature sequences found
 - short conserved sequences specific for a phylogenetically defined group of organisms
 - either complete or, more often, specific rRNA fragments can be compared
 - when comparing rRNA sequences between 2 organisms, their relatedness is represented by percent sequence homology
 - 70% is cutoff value for species definition

🌸 Endosymbiosis hypothesis

- Ancestral eukaryotic cell had lost cell wall
- Engulfment of an endosymbiote occurred - produced needed product such as ATP
- Genome reduction occurred
- Evolution of organelles
 - mitochondria
 - hydrogenosome
 - chloroplasts

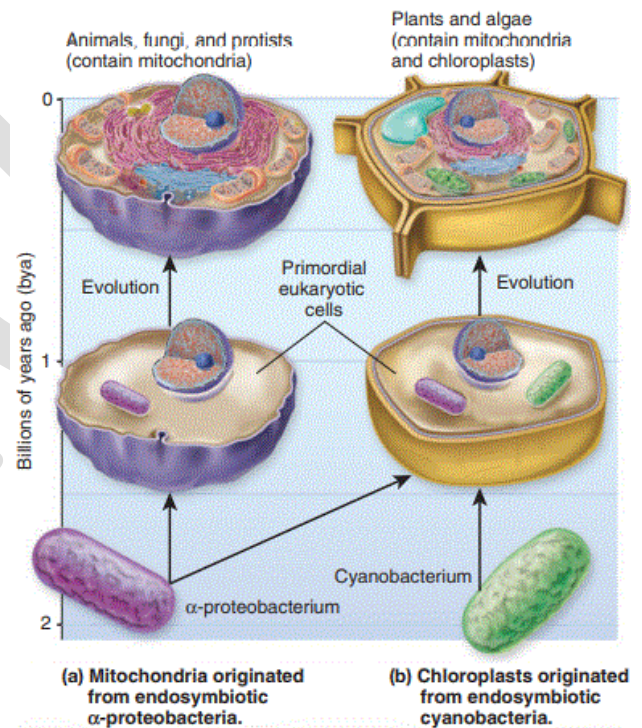


Figure 19.10 The Endosymbiotic Theory. (a) According to this hypothesis, mitochondria were derived from an α -proteobacterium. (b) A similar phenomenon occurred for chloroplasts, which were derived from cyanobacteria.

هاي النظرية بتحكي انو زمان كثير ال eukaryotic cell خسرت ال cell wall تبعها وبالتالي صارت قادرة تعمل endocytosis. هاي eukaryotic cell قدرت تبتلع نوعين من البكتيريا وما هضمتهم وما حطتهم لانو البكتيريا وفرت للخلية هاي الطاقة وبنفس الوقت الخلية وفرت غذاء للبكتيريا ولأنو الخلية "رح تسمحلها تضل" بس اذا وفرتها طاقة فصار في ضغط على البكتيريا بإنها بس لازم تحمل الجينات الي الها دخل بتحويل

الطاقة.. ومع مرور الوقت استمر ال genome reduction وبهيك تحولت هاي البكتيريا مع الزمن لل mitochondria الي كلنا منعرفها

والنوع الثاني من البكتيريا هو ال cyanobacteria الي قادرة على عمل البناء الضوئي.. هاي صارت بعدين chloroplast ولكل وحدة منهم في دراسات عليها بتدعمها

❖ *Bergey's Manual of Systematic Bacteriology*: a Detailed work containing descriptions of all bacterial species currently identified لازم تحفظوا اسم الكتاب

The End (~▽~)~ ~(▽~)~